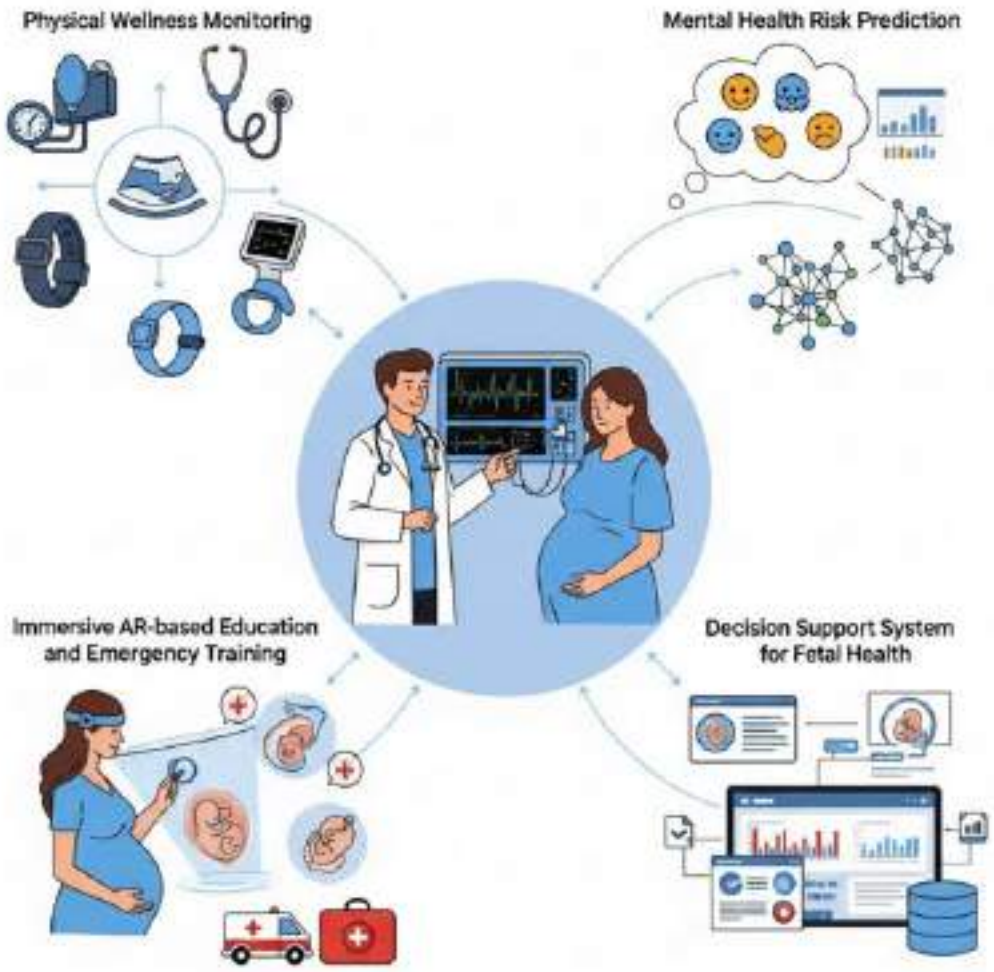
PREG ASSIST

Group :- 25-26J-253

Introduction



PREG ASSIST

Title of The Individual Component :

AR and ML Module for Personalized Pregnancy Emergency Training

Specialization : Information Technology

AR and ML Module for Personalized Pregnancy Emergency Training

Research Problem

Pregnant women and caregivers lack interactive, personalized emergency training

Existing methods are static, non-adaptive, and difficult to access when needed.

Online platforms risk privacy exposure for sensitive maternal health data.

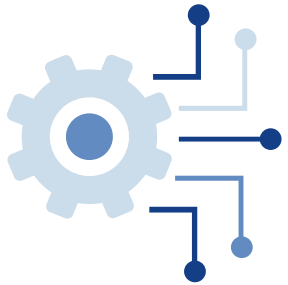
AR and ML Module for Personalized Pregnancy Emergency Training

Reasearch Gap and Solution

Existing Systems	Shortcomings in Existing Systems	Research Gap	How Proposed System Addresses It
Road to Birth – AR midwife training (generic, not personalized) [1]	• No user data adapted • Generic AR application	Lack of personalized emergency training	• Cross-domain transfer learning • Personalization
The Bump – Pregnancy tracking app (2D/static models, limited engagement) [2]	• Text/video/static AR • Low knowledge retention	Limited interactivity and engagement	• Interactive 3D AR modules • Step-by-step guidance • AR model compression
AR Pregnancy App – Cloud-based AR pregnancy lifestyle app (data stored on server) [3]	• Cloud-based processing • Data exposure risk	Privacy concerns with sensitive data	• Offline ML processing • Data security measures

AR and ML Module for Personalized Pregnancy Emergency Training

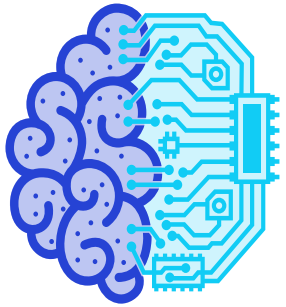
Application of Specialization based Knowledge



Main Domain	Latest Technologies Applied
Machine Learning	• Cross-domain transfer learning • TensorFlow Lite (on-device ML)
Augmented Reality	• AR model compression (LOD models + Polygon reduction) • ARCore/ARKit integration
Mobile Computing	• Hybrid mobile app design (offline ML + online AR) • Local data caching
Data Privacy & Security	• Offline processing • On-device storage

AR and ML Module for Personalized Pregnancy Emergency Training

Application of Specialization based Knowledge



How to evaluate?

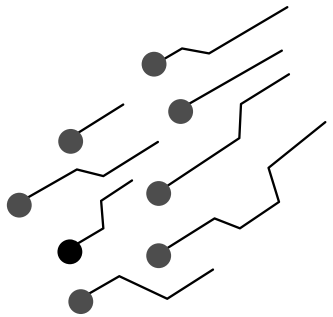
- ML model**:** Accuracy
- Personalization**:** Relevance of recommended AR modules
- User engagement**:** AR usability and learning retention
- System performance**:** AR loading speed and responsiveness

Data availabilty

- A custom dataset will be developed in collaboration with medical professionals.

Ethical Clearance

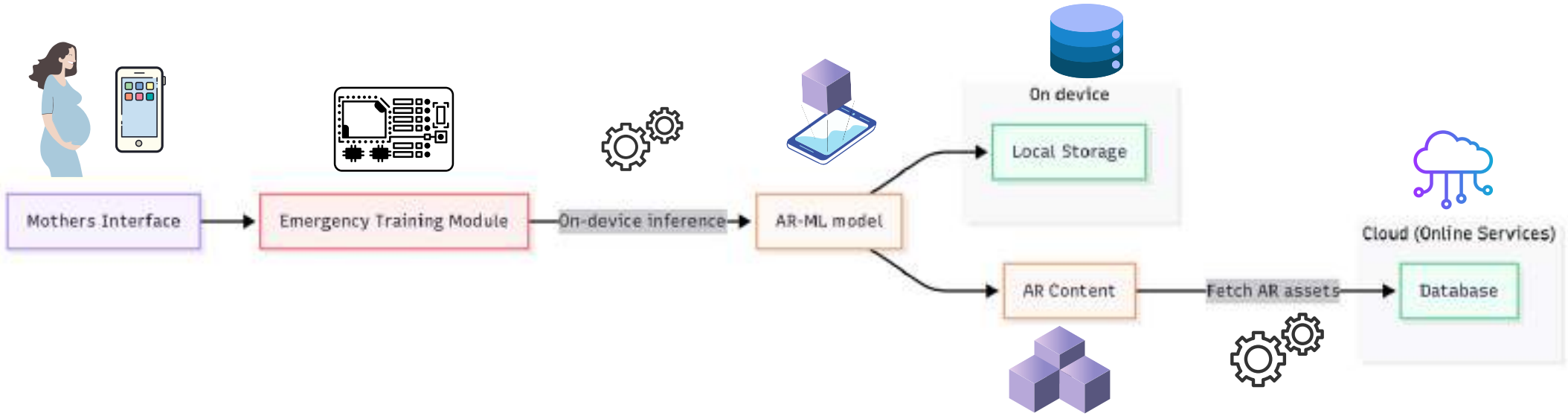
- User health data stored locally → ensures privacy
- Informed consent obtained before data collection
- Sensitive medical data handled securely
- AR recommendations support training only, not medical diagnosis



AR and ML Module for Personalized Pregnancy Emergency Training

Solution Implementation Details

System Overview of the Individual Component



How to use it in real-world scenario

- Mothers Log there health data via app, and receive a suitable emergency training for the week

AR and ML Module for Personalized Pregnancy Emergency Training

References

[1] Newcastle University, “Road to Birth: AR-based midwife training.” [Online]. Available: https://www.newcastle.edu.au/newsroom/featured/road-to-birth-delivers-world-first-virtual-window-into-pregnancy. [Accessed: Aug. 25, 2025].

[2] The Bump, “Pregnancy tracking with 3D models.” [Online]. Available: https://www.thebump.com/pregnancy-app. [Accessed: Aug. 25, 2025].

[3] AugmentedReality.Health, “AR Pregnancy App: Diet and lifestyle guidance.” [Online]. Available: https://augmentedreality.health/AR-Pregnancy-app.cfm. [Accessed: Aug. 25, 2025].

Title of The Individual Component -

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

Specialization : Information Technology

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

Research Problems

- How to predict pregnancy complications using a combination of parameters rather than single measures.
- How to continuously monitor multiple maternal health parameters to detect pregnancy risks early.
- How to provide personalized, real-time recommendations and alerts to mothers and doctors.

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

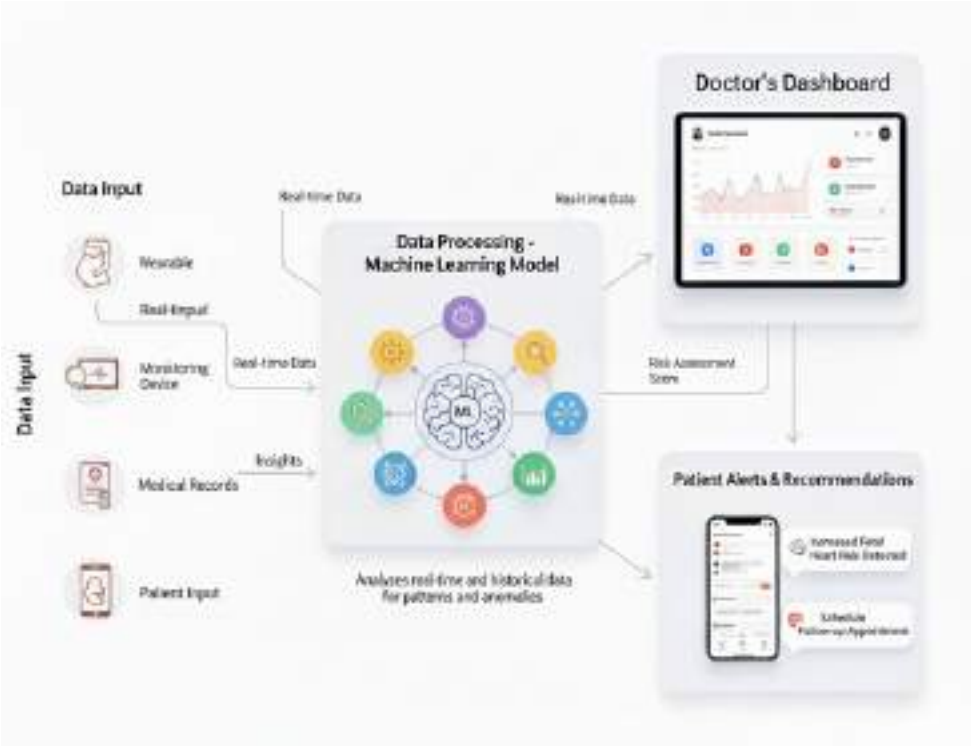
Research gap and Solutions

Existing Systems	Existing System Shortcomings	Research Gap	Proposed System Solution
Pregnancy+	• Tracks limited parameters (weight, BP) • Provides general tips only	No integration of multiple maternal parameters for early and accurate risk detection	Combine multiple maternal parameters into an ML model for accurate risk prediction with explainable outputs [1][2]
BabyCenter	• Provides weekly tips • Advice is not personalized • Not trimester-specific	Lack of personalized, risk-based recommendations	Generate trimester-specific, personalized health guidance based on predicted risk and contributing factors [3]
Fitbit	• Monitors vitals (HR, activity) • No explainable AI • No clinical risk analysis	Missing explainable prediction of complications	Use Random Forest with feature importance to provide explainable risk outputs for mothers and doctors [4]

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

Solution Implementation Details

System overview of the Individual Component



How to use it in a real-world scenario

- Mothers log health data via app or wearable and receive alerts and tips.

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

Application of Specialization-based Knowledge

Main domains

- Machine Learning for Maternal Risk Prediction
- Explainable AI (XAI) for Clinically Meaningful Alerts
- Wearable Device Integration for Data Collection

Special Technologies used in main domain

- Python → data preprocessing, feature engineering, and integration.
- Random Forest → build and train predictive model to classify maternal health risk as Normal / At Risk / High Risk.
- SHAP → explain why a specific risk alert was generated (clinically transparent AI).
- Dash → create the Visual Health Score Dashboard for mothers and doctors.
- Rule-Based Logic →personalized advice using medical guidelines
- Wearable Device APIs → collect real-time maternal parameters like heart rate, BP, temperature.
- Lazy Loading → manage continuous wearable + manual inputs smoothly.

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

Application of Specialization-based Knowledge

How to evaluate?

- Check if the system correctly detects complications and gives accurate alerts.
- Measure improvements in maternal and fetal health outcomes.

Data availabilty

Maternal Health Risk Assessment Dataset (Source: Mendeley Data).

Ethical Clearance

- Use anonymized clinical data.

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

References

1. Smith, J. et al., Mobile Health Applications in Pregnancy Monitoring, Journal of Medical Systems, 2021.
2. Davis, H., Pregnancy Apps and Maternal Health: A Review, Journal of Perinatal Medicine, 2020.
3. Thompson, R., AI for Predicting Pregnancy Complications, IEEE Access, 2022.
4. Miller, A., Prenatal Health Guidance Systems, Journal of Health Informatics, 2020.

Title of The Individual Component -AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women

Specialization : BSc. Hons in Information Technology Specializing in Infromation Technology

Research Problem

How can maternal mental health chatbots move beyond predefined responses to deliver adaptive generative AI guidance and alert trusted contacts in high-risk situations?

How can on-device emotion detection ensure privacy while adapting through federated learning without sending raw sensitive data to the cloud?

How can multimodal analysis (images, voice, and text) be effectively combined to generate accurate mental health risk predictions, instead of relying on single modal or limited dual modal systems?

AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women

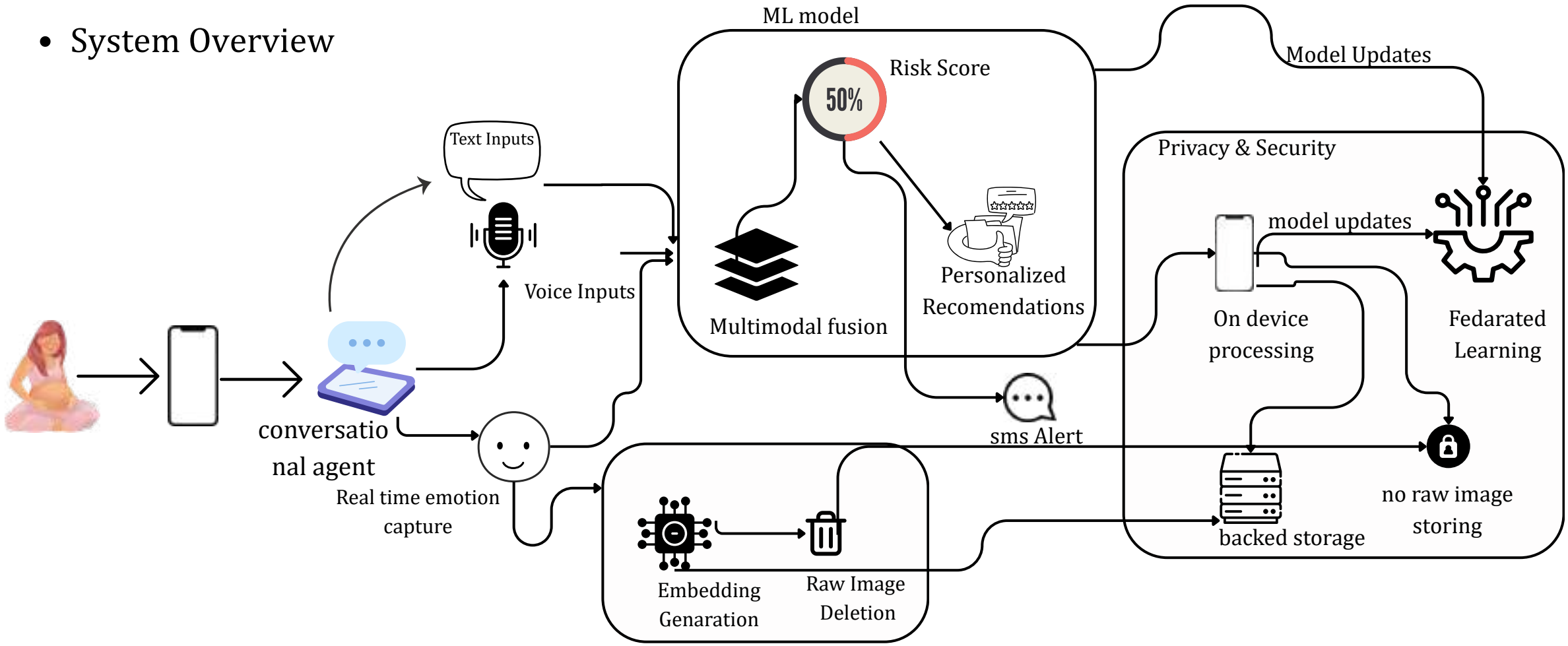
Reasearch Gap and System comparison

Existing Systems	Adaptive generative AI guidance + emergency alerts	Privacy (On-Device + Federated Learning)	Multimodal Fusion (Image + Voice + Text)
Wysa[1]	Predefined responses only	Cloud-based	Text-only
Replika[2]	✓ Generative AI, Alerts	No federated learning	Limited (mainly text/voice)
Woebot Health[3]	No personalization and No alerts	Cloud-based	Mostly text, lacks full multimodality
Proposed System	✓ Adaptive AI + emergency alerts	✓ On-Device + Federated Learning	✓ Full multimodal (image, voice, text)

AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women

Solution Implementation Details

- System Overview

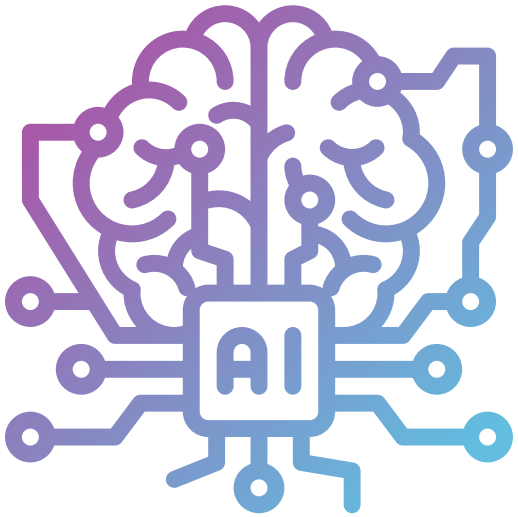


AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women

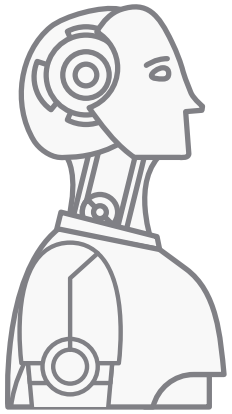
Application of Specialization-based Knowledge

Main Domains

- Artificial Intelligence(AI)
- Natural Language Processing(NLP)
- Edge/On-Device AI
- Federated Learning



Key Technologies



- Generative AI
- Multimodal Emotion Recognition
- On-device Processing (privacy-preserving analysis)
- Federated Learning
- Fusion Algorithm for Risk Scoring

Application of Specialization-based Knowledge

- How to Evaluate ?

  Model Evaluation - Train and test multimodal model on FER-2013 (images) + standard voice/text datasets.

  Metrics: Accuracy ≥80%, Precision/Recall ≥75%, Low false positives.

- Validation
  - Pilot testing with pregnant women (simulated scenarios).
  - Measure latency and reliability of real time mobile deployment.
  - Compare chatbot risk scores with expert/clinical assessments.

Application of Specialization-based Knowledge

- Data Availability
  - Image Detection: Use FER-2013 available on Kaggle.
- Ethical Clearance
  - No sensitive patient data will be collected.
  - Only publicly available datasets (FER-2013) are used for model training.

Application of Specialization-based Knowledge

Component How to use it in a real-world scenario

- Can support doctors and midwives with patient emotional insights during checkups.
- Integration with Healthcare Providers

References

[1] A. Martinez, L. Brown, J. Wang, and P. Singh, “Conversational Agents for Maternal Mental Health: A Review,” Journal of Medical Internet Research, vol. 26, 2024.

[2] Y. Chen, X. Li, M. Zhou, and T. Wu, “AFLEMP: Attention-Based Federated Learning for Multimodal Emotion Prediction,” IEEE Transactions on Affective Computing, 2025.

[3] R. Gupta, K. Sharma, D. Mehta, and P. Rao, “Multimodal Mental Health Monitoring: Challenges and Opportunities,” IEEE Access, vol. 12, pp. 7654–7665, 2024.

[4] L. Zhang, H. Kim, A. Torres, and M. Singh, “Advances in Multimodal Emotion Recognition for Healthcare Applications,” Frontiers in Artificial Intelligence, 2023.

[5] S. Patel, R. Verma, and A. Desai, “AI-Powered Maternal Health Interventions with IoT Integration,” Healthcare Technology Letters, vol. 12, no. 2, pp. 155–163, 2024.

Title of The Individual Component - **Decision Support System for Fetal Health Classification**

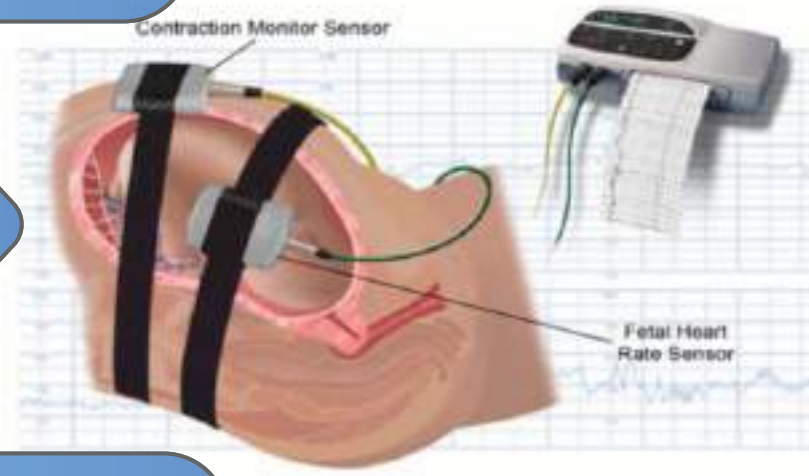
Specialization : Information Technology

Decision Support System for Fetal Health Classification

Research Problems

How to provide clinically meaningful explanations for model predictions?

Lack of confidence scoring in current models creates a significant safety concern. In real clinical environments



How to optimize machine learning models for on-device inference while maintaining accuracy and minimizing latency, storage, and power consumption.?

Decision Support System for Fetal Health Classification

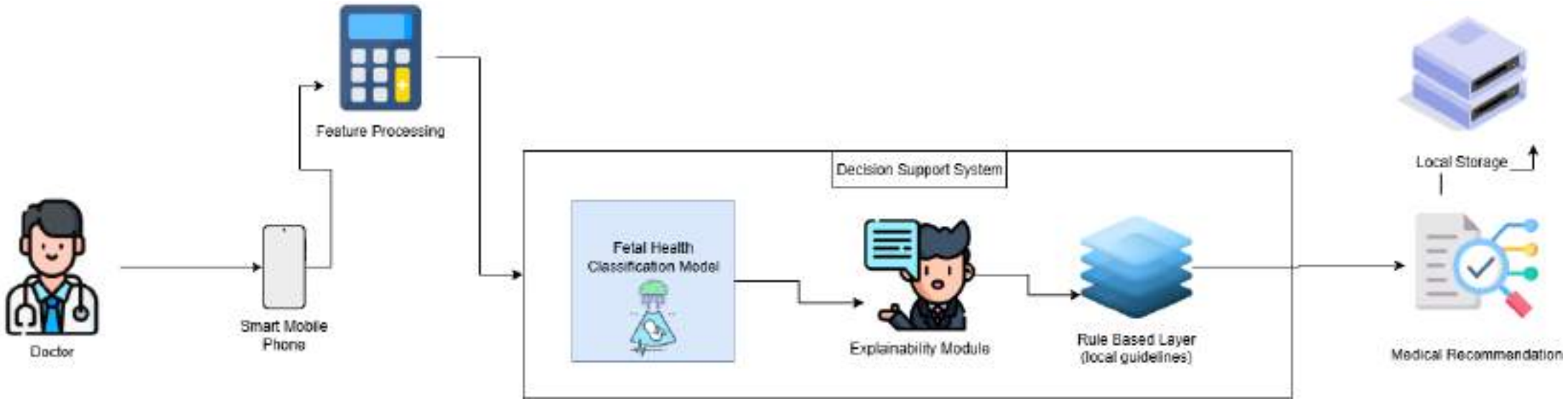
System Comparison

Existing Systems	Shortcomings in Existing Systems	Research Gap	How Proposed System Addresses It
PeriWatch Vigilance	• Require significant infrastructure [1] • High continuous costs [2]	• On device deployment	• Lightweight system • Use on device hardware • Offline Capability • Resource-efficient • Cost-efficient
Mushtaq & Veningston (2024)	• Lack clinically meaningful explanation [3] • Lack of confidence	• Explainability	• Clinically meaningful explanation • Confidence Score • Localized patient data

Decision Support System for Fetal Health Classification

Solution Implementation Details

System Overview of the Individual Component



How to use it in a real-world scenario

- Integration into Clinical Workflow
- Rural/Resource-Limited Settings

Decision Support System for Fetal Health Classification

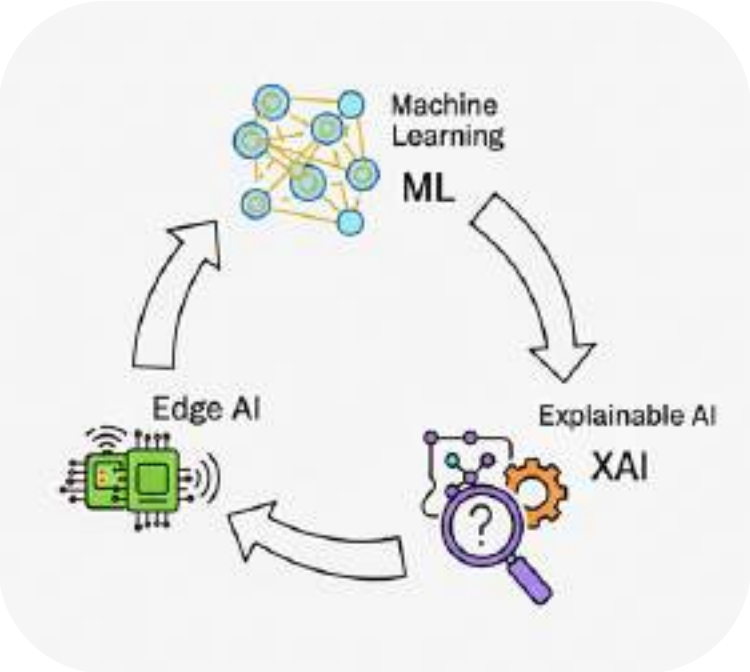
Application of Specialization-based Knowledge

Main Domains

- Machine Learning(ML)
- Explainable AI (XAI)
- Edge AI
- Mobile App Development

Specific Technologies used in the Main Domain

- Random Forest Technique
- SHapley Additive exPlanations (SHAP)
- TensorFlow Lite
- Lazy Loading
- Quantization
- Python
- Flutter



Decision Support System for Fetal Health Classification

Application of Specialization-based Knowledge

How to Evaluate ?

- Model Evaluation - Precision, Recall and F1-score metrics. / Confusion Matrix
- Confidence Aware Evaluation - Calibration curves (Reliability Diagrams)
- Explainability Validation - Perform expert review
- Deployment & Usability Validation - Measure model size & inference time

Data Availability ?

- Fetal Health Classification dataset available on Kaggle

Ethical Clearance ?

- public and anonymized dataset

Decision Support System for Fetal Health Classification

References

[1] PeriWatch Vigilance, Perigen. [Online]. Available: https://perigen.com/periwatch vigilance/ [Accessed: Jul. 21, 2025].

[2] S. Lavi, “PeriWatch Vigilance Pricing 2024 - Why 3.5/5?” ITQlick, Jun. 24, 2025. [Online]. Available: https://www.itqlick.com/periwatch-vigilance/pricing

[3] G. Mushtaq and V. K, "AI-driven Interpretable Deep Learning-based Fetal Health Classification," SLAS Technology, vol. 29, p. Article 100206, 2024.

PREG ASSIST

Commercialization

Invesment

- Server & Host Costing
- 3D assets cost
- Purchasing necessary software and equipments
- Product marketing cost
- Maintenance budget

How to recover the budget?

1. Identifying the Target Audience:
   - Pregnant Mothers
   - VOG Doctors
2. Revenue Generation:
   - Premium Features
3. Promotions:
   - Collaboration with Government and Private Hospitals
   - Social Media